Lesson 9: Flowchart Symbols And Program Logic

Generated Jul 4, 2026 5:08 PM

What We'll Learn

By the end of this lesson, we can: - Identify the basic flowchart symbols and their uses. - Use the correct symbol for Start, End, Process, Input/Output, Decision, and Flowline. - Trace a simple flowchart from start to end. - Explain the logic shown by a flowchart using our own words.

Words We'll Use

- Flowchart - a visual diagram that shows the steps of a process or program.
- Terminal - oval symbol for Start or End.
- Process - a rectangle used for actions, calculations, or assignments.
- Input/Output - a parallelogram used for entering data or showing results.
- Decision - a diamond used for questions with Yes/No or True/False answers.
- Flowline - an arrow that shows the direction of the logic.
- Program Logic - the order and reasoning behind how a program works.

Before We Start

Why might a drawing help us understand a program before writing the code?

Possible student ideas: - A drawing makes the steps easier to see. - It helps us plan before coding. - It helps us find missing steps. - It helps us explain the program to others.

Let's Understand

Flowchart symbol guide A flowchart is a visual plan of a program or process. Instead of writing code immediately, we first draw the logic using standard symbols.

Each symbol has a specific purpose: 1. Terminal / Oval - Used for the beginning and ending of a flowchart. Example: Start, End 2. Process / Rectangle - Used for actions, calculations, or assignments. Example: sum = num1 + num2 3. Input/Output / Parallelogram - Used when the program receives input or displays output. Example: Input name, Display total 4. Decision / Diamond - Used when the program asks a question with two possible answers. Example: grade >= 75? 5. Flowline / Arrow Used to show which step happens next.

Important rules: - A flowchart should begin with Start. - A flowchart should end with End. - Arrows should clearly show the direction of the logic. - Use the correct symbol for each step. - A decision symbol should have labeled branches such as Yes/No or True/False. - The flowchart should be easy to trace from beginning to end.

Let's Look at Examples

Example 1: Sequence - Display a Welcome Message

A simple program displays a welcome message to the user. This example shows a straight sequence from Start to End.

> **Common mistake**
>
> Do not use a diamond here because the program is not asking a Yes/No question.

> **Mini-check**
>
> Which shape is used for displaying the message?

Example 2: Input and Output - Greet the User

The program asks for the user's name, then displays a greeting using that name.

> **Common mistake**
>
> Do not place 'Input name' inside a rectangle. Input and output steps should use a parallelogram.

> **Mini-check**
>
> Why is 'Input name' written inside a parallelogram?

Example 3: Process - Add Two Numbers

The program receives two numbers, computes their sum, and displays the result.

> **Common mistake**
>
> Do not use a parallelogram for 'sum = num1 + num2' because it is not receiving or displaying data. It is calculating a value.

> **Mini-check**
>
> Which step creates a new value?

Example 4: Decision - Check if a Student Passed

The program receives a grade. If the grade is 75 or higher, it displays Passed. Otherwise, it displays Try Again.

> **Common mistake**
>
> Do not forget to label the two arrows from the diamond. Without Yes and No labels, the reader may not know which path to follow.

> **Mini-check**
>
> What are the two possible answers to the decision 'grade >= 75?'

Example 5: Complete Logic - Compute Average and Check Result

The program receives quiz, exam, and activity scores. It computes the average, then checks if the average is passing.

> **Common mistake**
>
> Do not check 'average >= 75?' before computing the average. The value must be created first before it can be tested.

> **Mini-check**
>
> Why should the average be computed before the decision diamond?

Let's Practice

Practice 1: Identify the Symbol

Write the correct flowchart symbol for each item.

1. Start
2. Input age
3. total = price * quantity
4. Display "Welcome"
5. age >= 18?
6. End
7. Arrow showing the next step

Practice 2: Match the Symbol to Its Use

Match Column A with Column B.

Column A: 1. Oval 2. Rectangle 3. Parallelogram 4. Diamond 5. Arrow

Column B: A. Shows direction B. Used for decision C. Used for Start and End D. Used for input and output E. Used for process or calculation

Now We Apply

Create a simple flowchart for this problem:

A program asks for a student’s name and grade, then displays the name and grade.

Required steps:

1. Start
2. Input student name
3. Input grade
4. Display student name
5. Display grade
6. End

Students should draw the flowchart using the correct symbols.

Challenge Task

Create a flowchart for this problem:

A program asks for the price of an item and the quantity bought. It computes the total amount and displays the total.

Challenge question: Which step uses a rectangle? Why?

How Our Work Will Be Checked

Your work will be checked using these criteria:

Criteria	Excellent	Good	Needs Improvement
Correct Symbols	All symbols are used correctly	Most symbols are correct	Many symbols are incorrect
Complete Flow	Has Start, steps, and End	Missing one minor part	Missing important parts
Arrows and Direction	Arrows are clear and connected	Some arrows need improvement	Arrows are confusing or missing
Logic	Steps are in the correct order	Few steps are slightly unclear	Logic is hard to follow
Explanation	Can explain the flow clearly	Explanation is partly clear	Cannot explain the flow

Let's Reflect

Answer these questions: 1. Why is it important to use the correct flowchart symbol? 2. Which symbol is easiest for you to remember? 3. Which symbol is most confusing for you? 4. How can flowcharts help us before coding? 5. What mistake should you avoid when making a flowchart?

How We Show Learning

tudents will show learning by: - Identifying flowchart symbols correctly. - Drawing a simple flowchart. - Tracing a flowchart from Start to End. - Explaining what each symbol means. - Explaining the logic of their flowchart.

Student Activities

Symbol Sorting Drill. Wrong Shape Correction. Mini Flow Trace. Symbol Choice Explanation. Shape Memory Exit Ticket. Match each symbol to its purpose. Correct wrong-symbol examples. Explain the shape choice for five program steps.

Resources / References

JDK, IDE, projector or screenshots, printed worksheets, notebook, sample code snippets, and teacher-created answer guides.